Featured in Development

Seven Operators to Get Started with RxJS

If you're just getting started with Reactive JavaScript and RxJS, the list of available operators can be overwhelming. If you're just getting started, which ones do you actually need? In this article, Vinvent Tunru introduces sven operators along with examples that will help you get a feel for when each of these operators could come in handy.

- Case Studies
- Microservices
- Patterns
- Security

Featured in Architecture & Design

Roundtable: The Role of Enterprise Architecture in a Cloudy World


Do enterprise architects still matter? Has a cloud-native development model—not to mention a DevOps and SRE approaches to operations—fundamentally changed how we think about enterprise architecture? In this roundtable, we talk to four experienced architects to find the answer.

All in **Architecture & Design**

- Data Science
  - Big Data
  - Machine Learning
  - NoSQL
  - Database
  - Data Analytics
  - Streaming

Featured in Data Science

Big Data Infrastructure @ LinkedIn


Shirshanka Das describes LinkedIn's Big Data Infrastructure and its evolution through the years, including details on the motivation and architecture of Gobblin, Pinot and WhereHows.

All in **Data Science**

- Culture & Methods
  - Agile
  - Leadership
  - Team Collaboration
  - Testing
  - Project Management
  - UX
  - Scrum

- Lean/Kanban
- Personal Growth

89
Shares

Featured in Culture & Methods

Roundtable: The Role of Enterprise Architecture in a Cloudy World

60


Do enterprise architects still matter? Has a cloud-native development model—not to mention a DevOps and SRE approaches to operations—fundamentally changed how we think about enterprise architecture? In this roundtable, we talk to four experienced architects to find the answer.

All in **Culture & Methods**

DevOps

- Infrastructure
- Continuous Delivery
- Automation
- Containers
- Cloud

Featured in DevOps

Roundtable: The Role of Enterprise Architecture in a Cloudy World


Do enterprise architects still matter? Has a cloud-native development model—not to mention a DevOps and SRE approaches to operations—fundamentally changed how we think about enterprise architecture? In this roundtable, we talk to four experienced architects to find the answer.

All in **DevOps**

- Podcasts

- Streaming
- Machine Learning
- Reactive

Wiring Microservices with Spring Cloud

d by Rob Harrop on Jun 15, 2016. Estimated reading time: 18 minutes / 6 Discuss

Key takeaways

- Spring Cloud provides a wealth of options for wiring service dependencies in microservice systems.
- Spring Cloud Config provides Git-managed versioning for configuration data and enables dynamic refresh of this data without the need for restart.
- Pairing Spring Cloud with the Netflix Eureka and Ribbon components allows application services to find each other dynamically, and pushes load-balancing decisions away from a dedicated proxying load-balancer and into the client services.
- Load balancing solutions like AWS ELB still have a place at the edge of our system where we don't control the inbound traffic.
- For communication between middle-tier microservices, Ribbon provides a more reliable and more performant solution that doesn't couple you to any particular cloud provider.

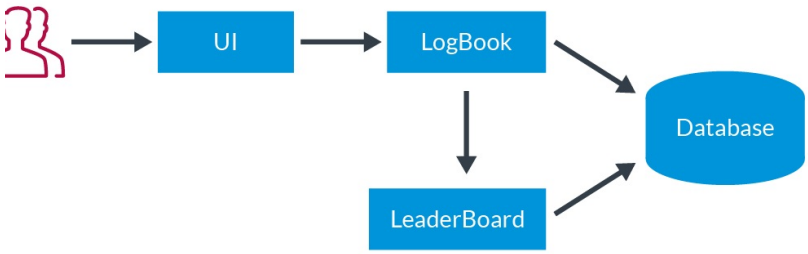
Introduction

As we move towards microservice-based architectures, we're faced with an important decision: how do we wire our services together? Components in a monolithic system communicate through a simple method call, but components in a microservice system likely communicate over the network through REST, web services or some RPC-like mechanism.

In a monolith we can avoid issues of service wiring altogether and have each component create its own dependencies as it needs them. In reality, we rarely do this. Such close coupling between a component and its dependencies makes our system overly rigid, and hampers our testing efforts. Instead, we opt to externalise a component's dependencies and inject them when the component is created; dependency injection is service wiring for classes and objects.

Suppose that we decide to implement an application as a set of microservices, we have wiring options similar to those we have for a monolith. We can hard-code the addresses of our dependencies, tying our services together closely. Alternatively, we can externalise the addresses of the services we depend on, and supply them at deploy time or run time. In this article, we'll explore how each of these options manifests in a microservices application with Spring Boot and Spring Cloud.

Consider the simple `repmax` microservice system shown below:



The repmax system

The repmax application tracks a user's weight-lifting history and tracks the top five users for each lift as the leaderboard. The `logbook` service receives workout data from the UI and stores the full history for each user. When a user logs a lift in a workout, the `logbook` sends the details for that lift to the `leaderboard` service.

From the diagram, we see that the `logbook` service depends on the `leaderboard` service. Following good practice, we abstract this dependency behind an interface, the `LeaderBoardApi`:

```
public interface LeaderBoardApi {

    void recordLift(Lift lift);
}
```

Since this is a Spring application, we'll be using `RestTemplate` to handle the details of the communication between the logbook and leaderboard services:

```
abstract class AbstractLeaderBoardApi implements LeaderBoardApi {

    private final RestTemplate restTemplate;

    public AbstractLeaderBoardApi() {
        RestTemplate restTemplate = new RestTemplate();
        restTemplate.getMessageConverters().add(new FormHttpMessageConverter());
        this.restTemplate = restTemplate;
    }

    @Override
    public final void recordLift(Lifter lifter, Lift lift) {
        URI url = URI.create(String.format("%s/lifts", getLeaderBoardAddress()));

        MultiValueMap<String, String> params = new LinkedMultiValueMap<>();
        params.set("exerciseName", lift.getDescription());
        params.set("lifterName", lifter.getFullName());
        params.set("reps", Integer.toString(lift.getReps()));
        params.set("weight", Double.toString(lift.getWeight()));

        this.restTemplate.postForLocation(url, params);
    }

    protected abstract String getLeaderBoardAddress();
}
```

The `AbstractLeaderBoardApi` class captures all the logic needed to package up a `POST` request to the `leaderboard` service, leaving subclasses to specify the exact address for the `leaderboard` service.

The simplest possible model for wiring one microservice to another is to hard-code the address of each dependency that a service needs. This corresponds to hard-coding the instantiation of a dependency in the monolith world. This is easily captured in the `StaticWiredLeaderBoardApi` class:

```
public class StaticWiredLeaderBoardApi extends AbstractLeaderBoardApi {

    @Override
    protected String getLeaderBoardAddress() {
        return "http://localhost:8082";
    }
}
```

The hard-coded address for the service allows us to get started quickly, but is not tenable in a real environment. Every different deployment of the services requires custom compilation, this quickly becomes painful and error-prone.

If this were a monolith and we were looking to refactor our application to remove the hard-coded address, we'd start by externalising the address into some configuration file. The same approach works for our microservice application: we push the address into a configuration file and have our API

nentation read the address from the configuration.

Boot makes defining and injecting configuration parameters trivial. We add the address parameter to the `application.properties` file:

```
rboard.url=http://localhost:8082
```

n inject this parameter into our `ConfigurableLeaderBoardApi` implementation using the `@Value` annotation:

```
60
: class ConfigurableLeaderBoardApi extends AbstractLeaderBoardApi {

rivate final String leaderBoardAddress;

Autowired
ublic ConfigurableLeaderBoardApi(@Value("${leaderboard.url}") String leaderBoardAddress) {
    this.leaderBoardAddress = leaderBoardAddress;


Override
rotected String getLeaderBoardAddress() {
    return this.leaderBoardAddress;
}
}
```

The Externalized Configuration support in Spring Boot allows us to change the value of `leaderboard.url` not just by editing the configuration file, but also by specifying an environment variable when starting our application:

```
LEADERBOARD_URL=http://repmax.skipjaq.com/leaderboard java -jar repmax-logbook-1.0.0-RELEASE.jar
```

We can now point an instance of the `logbook` service to any instance of the `leaderboard` service without having to make code changes. If we are following 12 factor principles in our system, then the wiring information will likely be available in the environment, so it can be mapped directly into the application without too much fuss.

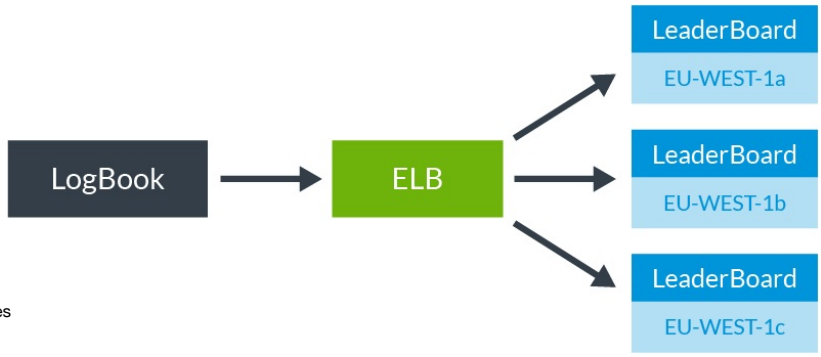
Platform-as-a-Service (PaaS) systems such as Cloud Foundry and Heroku expose wiring information for managed services such as databases and messaging systems through the environment, allowing us to wire in these dependencies in exactly the same way. Indeed, it makes little sense to differentiate between wiring two services together and wiring a service to its datastore; in both cases we are simply connecting two distributed systems.

Beyond Point-to-Point Wiring

For simple applications, external configuration for dependency addresses may well be sufficient. For applications of any size though, it's likely that we'll want to move beyond simple point-to-point wiring and introduce some form of load-balancing.

If each of our services depends directly on a single instance of its downstream services, then any failure in the downstream chain is likely to be catastrophic for our end users. Likewise, if a downstream service becomes overloaded, then our users pay the price for this through increased response times. What we need is load balancing.

Instead of depending directly on a downstream instance, we want to share the load across a set of downstream service instances. If one of these instances fails or becomes overloaded then the other instances can pick up the slack. The simplest way to introduce load balancing into this architecture is using a load-balancing proxy. The diagram below shows how this might look in an Amazon Web Services deployment using Elastic Load Balancing:



r the Leader Board

60

than have the `logbook` service talk directly to the `leaderboard` service, each request is routed through an ELB. The ELB routes each request to the backend `leaderboard` services. With the ELB acting as an intermediary, load is shared across multiple leaderboard instances, helping to the load on individual instances.

balancing with ELB is dynamic; new instances can be added to set of backend instances at runtime so if we encounter a spike in incoming we can start more `leaderboard` instances to handle this spike.

Boot applications using the actuator expose a `/health` endpoint that the ELB monitors periodically. Instances that respond to these checks remain in the ELB's active set, but after a number of failed checks, the instance is removed from service.

The `leaderboard` service isn't the only service in our system that can benefit from this load-balancing. The `logbook` service, and indeed the front-end UI, both benefit from the scalability and resilience afforded by load balancing.

Dynamic Re-Configuration

Whether we are using AWS ELB, Google Compute Load Balancing or even our own load-balancing proxy using something like HAProxy or NGINX, we still need to wire our services up to the load balancer.

One approach to this is to give each load balancer a well-known DNS name, say `leaderboard.repmax.local` that can be hard-coded into the application using the static wiring approach shown earlier. This approach is reasonably flexible thanks in large part to how flexible DNS is. However, relying on a hard-coded name means we have to configure a DNS server in every environment we are running our services in. Requiring a customised DNS is particularly painful in development where we have to support many different OSes. A better solution is to inject whatever address the load balancer has naturally into our services as we did with `leaderboard.url` in the previous example.

In cloud environments such as AWS and GCP, load balancers - and their addresses - are ephemeral. When a load balancer gets destroyed and re-created it typically gets a new address. If we hard-code the address of the load balancer, we need to re-compile our code just to pick up the address change. With externalised configuration, we simply modify the configuration file and restart.

DNS is a handy way of hiding the ephemeral nature of load balancer addresses. Each load balancer is assigned a stable DNS name and it's this name that's injected into the calling service. When the load balancer is re-created, the DNS name is remapped to the new address for the load balancer. This DNS-based approach works well if you're prepared to run a DNS server in your environment. If you want to avoid running DNS but still allow dynamic re-configuration of your load balancers then Spring Cloud Config is the answer.

Spring Cloud Config runs a small service, the Config Server, to provide centralised access to configuration data through a REST API. By default, configuration data is stored in a Git repository and is exposed to our Spring Boot services through the standard `PropertySource` abstraction. Thanks to the use of `PropertySource`, we can seamlessly combine configuration in local properties files with configuration stored in the Config Server. For local development, we'll likely use configuration from a local properties file and only override this when deploying the application in a real environment.

To replace our `ConfigurableLeaderBoardApi` with an implementation that uses Spring Cloud Config we start by initialising a Git repo with the desired configuration:

```
mkdir -p ~/dev/repmax-config-repo
cd ~/dev/repmax-config-repo
git init
echo 'leaderboard.lb.url=http://some.lb.address' >> repmax.properties
git add repmax.properties
git commit -m 'LB config for the leaderboard service'
```

The `repmax.properties` file contains the configuration for the `default` profile of the `repmax` application. If we want to add configuration for another profile, say `development`, then we simply commit another file called `repmax-development.properties`.

To run the Config Server, we can either run the default Config Server provided by the `spring-cloud-config-server` project or we can create our own simple Spring Boot project that hosts the Config Server:

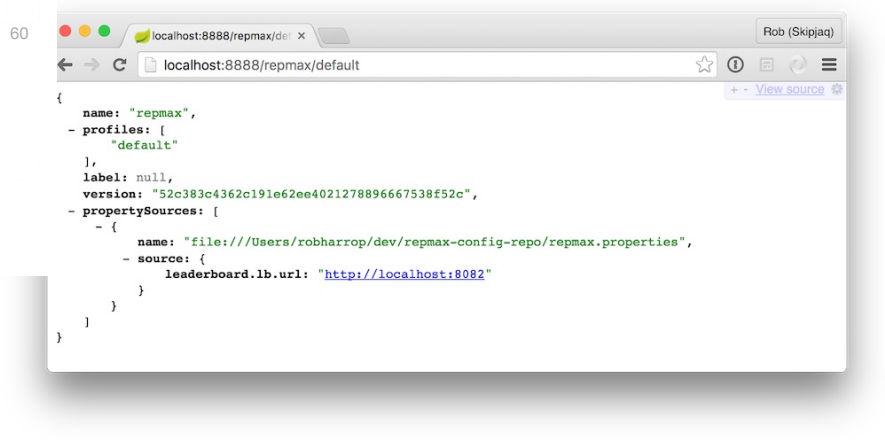
```
@SpringBootApplication
@EnableConfigServer
public class RepmaxConfigServerApplication {

    public static void main(String[] args) {
        SpringApplication.run(RepmaxConfigServerApplication.class, args);
    }
}
```

The `@EnableConfigServer` annotation starts Config Server in this tiny Spring Boot application. We point the Config Server at our Git repo using the `spring.cloud.config.server.git.uri` property. For local testing it makes sense to add this to the `application.properties` file for the Config Server application:


```
g.cloud.config.server.git.uri=file://${user.home}/dev/repmax-config-repo
```

way, every developer in our team can start up a Config Server on their machine and test it against a local Git repository. We can verify that the ties for the `repmax` application are exposed by Config Server by visiting `http://localhost:8888/repmax/default` in the browser when the Server is running:



Browsing configuration in Config Server

Here we can see that the `leaderboard.lb.url` property is exposed from the `repmax.properties` file and it has the value `http://localhost:8083`. The `version` property in the JSON payload shows which Git rev the config was loaded from.

At production time we take advantage of the `PropertySource` abstraction to supply the Git repository name as an environment variable:

```
SPRING_CLOUD_CONFIG_SERVER_GIT_URI=https://gitlab.com/rdh/repmax-config-repo java -jar repmax-config-server-1.0.0-RELEASE.jar
```

A Spring Cloud Config Client

Modifying the logbook service to read its configuration from our new Config Server requires only a few steps. First we add a dependency on `spring-cloud-starter-config` in the `build.gradle` file;

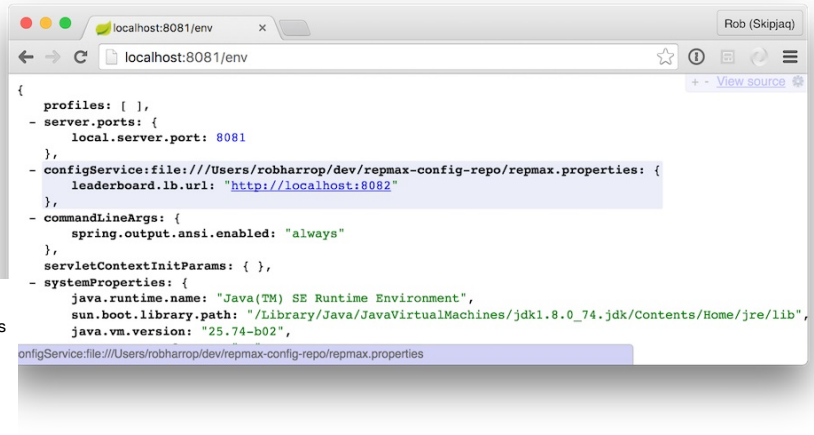
```
compile("org.springframework.cloud:spring-cloud-starter-config:1.1.1.BUILD-SNAPSHOT")
```

Next, we supply some basic bootstrap configuration needed by the Config Client. Recall that our Config Server exposes configuration from a file called `repmax.properties`. We need to tell the Config Client the name of our application. Such bootstrap configuration goes in the `bootstrap.properties` file of the `logbook` service:

```
spring.application.name=repmax
```

By default, the Config Client looks for a Config Server at `http://localhost:8888`. To tweak this, specify the `SPRING_CLOUD_CONFIG_URI` environment when starting the client application.

Once the client, in this case `logbook` is started, we can check that the configuration from the Config Server is correctly loaded by visiting `http://localhost:8081/env`:



ng that Config Client can see the Config Server

he `logbook` service configured to use Config Client, we can modify the `ConfigurableLeaderBoardApi` to obtain the load balancer address from `aderboard.lb.url` property exposed in the Config Server.

ling Dynamic Refresh

he configuration stored in a centralised place we have an easy way to change the `repmax` configuration in a way that is visible to all services. ver, picking up those configurations still requires a restart. We can do better. Spring Boot provides the `@ConfigurationProperties` annotation lows us to map configuration directly on to JavaBeans. Spring Cloud Config goes a step further, and exposes a `/refresh` endpoint in every client service. Beans that are annotated with `@ConfigurationProperties` have their properties updated whenever a refresh is triggered through the `/refresh` endpoint.

Any bean can be annotated with `@ConfigurationProperties`, but it makes sense to restrict refresh support to just the beans that contain configuration data. To this end, we extract a `LeaderboardConfig` bean that serves as a holder for the `leaderboard` address:

```
@ConfigurationProperties("leaderboard.lb")
public class LeaderboardConfig {

    private volatile String url;

    public String getUrl() {
        return this.url;
    }

    public void setUrl(String url) {
        this.url = url;
    }
}
```

The value of the `@ConfigurationProperties` annotation is the prefix for configuration values we want to map into our bean. Then, each value is mapped using standard JavaBean naming conventions. In this case, the `url` bean property is mapped to `leaderboard.lb.url` in the configuration.

We then modify `ConfigurableLeaderBoardApi` to accept an instance of `LeaderboardConfig` rather than the raw `leaderboard` address:

```
public class ConfigurableLeaderBoardApi extends AbstractLeaderBoardApi {

    private final LeaderboardConfig config;

    @Autowired
    public ConfigurableLeaderBoardApi(LeaderboardConfig config) {
        this.config = config;
    }

    @Override
    protected String getLeaderBoardAddress() {
        return this.config.getLeaderboardAddress();
    }
}
```

To trigger a config refresh, send an HTTP `POST` request to the `/refresh` endpoint of the `logbook` service:

```
curl -X POST http://localhost:8081/refresh
```

Towards Service Discovery

With Spring Cloud Config and a load-balancing proxy between our `logbook` and `leaderboard` services, our application is in good shape. However, there are still some improvements to be made.

If we're deploying in AWS or GCP, we can take advantage of the elasticity of the load balancers in those environments, but if we're using an off-the-shelf load balancing proxy like HAProxy or NGINX, then we are forced to handle service discovery and registration ourselves. Every new instance of the `leaderboard` has to be configured with the proxy and every failed instance has to be removed from the proxy. What we really want is dynamic discovery where each service instance registers itself ready for discovery by its consumers.

There's another issue lurking with the use of load-balancing proxies: reliability. The need to route all traffic through the proxy puts our system at the mercy of that proxy's reliability. Downtime in the proxy is going to cause downtime in the system. We might also wonder at the overhead of having to talk from client to proxy and from proxy to server.

To overcome these problems, Netflix created Eureka. Eureka is a client-server system providing service registration and discovery. As service instances start they register themselves with the Eureka server. Client services, such as our `logbook`, contact the Eureka Server to obtain a list of available services. Communication between clients and servers is point-to-point.

ı removes the need for a proxy, improving the reliability of our systems. If our `leaderboard` proxy dies, then the `logbook` service can no longer the `leaderboard` service at all. With Eureka in place, `logbook` knows about all of the `leaderboard` instances so, even if one fails, `logbook` can ove on to the next `leaderboard` instance and try again.

ight wonder whether the Eureka Server itself becomes a point of failure in our system architecture. Aside from the ability to configure a *cluster* eka Servers, each Eureka Client caches the state of the running services locally. Provided we have a service monitor such as `systemd` running reka Server, we can happily tolerate the occasional crash.

h Config Server, we can run Eureka Server as a small Spring Boot application:

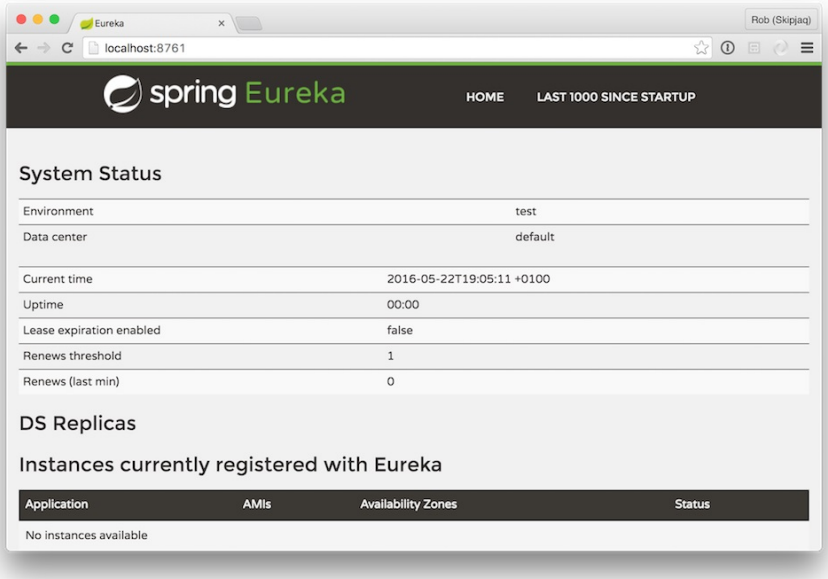
```
ıgBootApplication
leEurekaServer
: class RepmaxEurekaServerApplication {

ıblic static void main(String[] args) {
   SpringApplication.run(RepmaxEurekaServerApplication.class, args);
}
}
```

The `@EnableEurekaServer` annotation instructs Spring Boot to start the Eureka when the application starts. By default, the server will attempt to contact other servers for HA purposes. In a standalone installation it makes sense to turn this off. In `application.yml`:

```
server:
  port: 8761
eureka:
  instance:
    hostname: localhost
  client:
    registerWithEureka: false
    fetchRegistry: false
```

Note that we follow the common convention and run Eureka Server on port `8761`. Visiting `http://localhost:8761` shows the Eureka dashboard. Since we haven't registered any services yet, the list of available instance is empty:

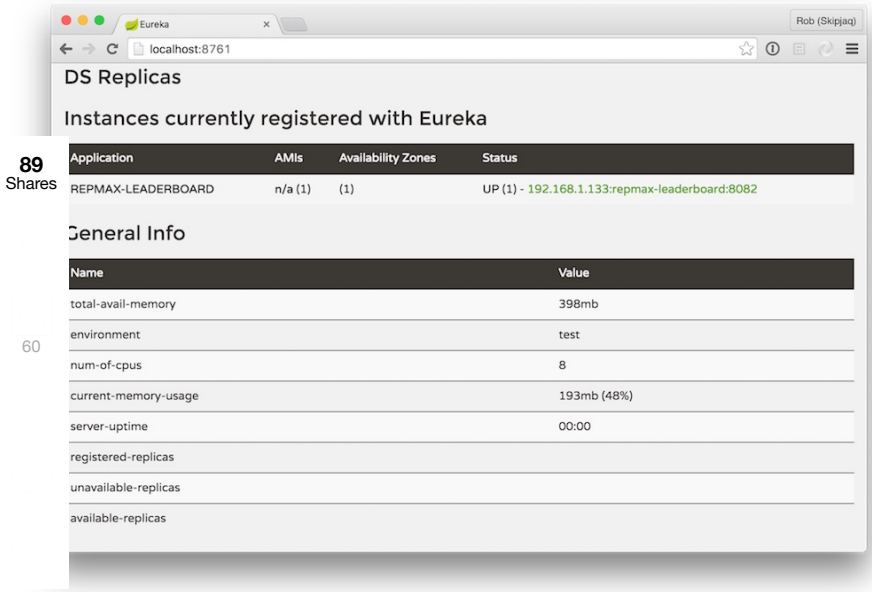


Blank Eureka dashboard

To register the `leaderboard` service with Eureka we annotate the application class with `@EnableEurekaClient`. We need to tell the Eureka Client where to find the server and what name to use for the application when registering it with the server. In `application.properties`:

```
spring.application.name=repmax-leaderboard
eureka.client.serviceUrl.defaultZone=http://localhost:8761/eureka
```

When the `leaderboard` service starts, Spring Boot detects the `@EnableEurekaClient` annotation and starts the Eureka Client that, in turn, registers the `leaderboard` service with the Eureka Server. The Eureka dashboard shows the newly registered service:



Eureka dashboard after registration

The `logbook` service is configured as a Eureka client in exactly the same way as the `leaderboard` service. We add the `@EnableEurekaClient` annotation and configure the Eureka Service URL.

With the Eureka Client enabled in the `logbook` service, Spring Cloud exposes a `DiscoveryClient` bean that allows us to lookup service instances:

```
@Component
public class DiscoveryLeaderBoardApi extends AbstractLeaderBoardApi {

    public DiscoveryLeaderBoardApi(DiscoveryClient discoveryClient) {
        this.discoveryClient = discoveryClient;
    }

    private final DiscoveryClient discoveryClient;

    @Override
    protected String getLeaderBoardAddress() {
        List<ServiceInstance> instances = this.discoveryClient.getInstances("repmax-leaderboard");
        if(instances != null && !instances.isEmpty()) {
            ServiceInstance serviceInstance = instances.get(0);
            return String.format("http://%s:%d", serviceInstance.getHost(), serviceInstance.getPort());
        }
        throw new IllegalStateException("Unable to locate a leaderboard service");
    }
}
```

We call `DiscoveryClient.getInstances` to obtain a list of `ServiceInstances`, each one corresponding to an instance of the `leaderboard` service that is registered with the Eureka Server. For simplicity, we pick the first one in the list and use that for our remote call.

Load Balancing on the Client

With Eureka in place, services now discover each other dynamically and communicate directly with each other, avoiding the overhead and possible point-of-failure that a proxying load balancer introduces. The trade-off, of course, is that we've pushed the complexity of load balancing into our code.

You'll notice that our `DiscoveryLeaderBoardApi.getLeaderBoardAddress` method naively selects the first `ServiceInstance` it finds for every remote call. This is hardly balancing the load across the available instances. Thankfully, there's another Netflix Cloud component available that can handle this client-side load balancing for us: Ribbon.

Using Ribbon with Spring Cloud and our existing Eureka setup is trivial. We simply add a dependency on `spring-cloud-starter-ribbon` in the `logbook` service and switch from using the `DiscoveryClient` to the `LoadBalancerClient`:

```
public class RibbonLeaderBoardApi extends AbstractLeaderBoardApi {

    private final LoadBalancerClient loadBalancerClient;
```

```
@Autowired
public RibbonLeaderBoardApi(LoadBalancerClient loadBalancerClient) {
    this.loadBalancerClient = loadBalancerClient;
}

@Override
protected String getLeaderBoardAddress() {
    ServiceInstance serviceInstance = this.loadBalancerClient.choose("repmax-leaderboard");
    if (serviceInstance != null) {
        return String.format("http://%s:%d", serviceInstance.getHost(), serviceInstance.getPort());
    } else {
        throw new IllegalStateException("Unable to locate a leaderboard service");
    }
```


he job of a choosing a `ServiceInstance` is passed off to Ribbon which has all the smarts for monitoring endpoint health and load balancing

ımary

the course of this article we've looked at a variety of approaches for wiring microservices together. The simplest possible approach is to hard-ıe address for each dependency that a service needs. This approach allows us to get started quickly, but is not tenable in a real environment.

ɔasic real-world application external configuration using an `application.properties` file for dependency addresses may well be sufficient. m-as-a-Service (PaaS) systems such as Cloud Foundry and Heroku expose wiring information allowing us to wire in these dependencies in / the same way.

applications however are likely to need to move beyond simple point-to-point wiring and introduce some form of load-balancing. Spring Cloud coupled with a load-balancing proxy is one solution, but if we're using an off-the-shelf load-balancing proxy like HAProxy or NGINX, then we are forced to handle service discovery and registration ourselves, and the proxy gives us a single point of failure for all traffic. Adding Netflix's Eureka and Ribbon components allows services in our applications to find each other dynamically and pushes load-balancing decisions away from a dedicated proxying load-balancer and in to the client services.

Load balancing solutions like AWS ELB still have a place at the edge of our system where we don't control the inbound traffic For communication between middle-tier microservices, Ribbon provides a more reliable and more performant solution that doesn't couple you to any particular cloud provider.

About the Author

Rob Harrop is CTO at Skipjaq, applying machine learning to the problem of performance management. Before Skipjaq, Rob was most well known as a co-founder of SpringSource, the software company behind the wildly-successful Spring Framework. At SpringSource he was a core contributor to the Spring Framework and led the team that built dm Server (now Eclipse Virgo). Prior to SpringSource, Rob was (at the age of 19) co-founder and CTO at Cake Solutions, a boutique consultancy in Manchester, UK. A respected author, speaker and teacher, Rob writes and talks frequently about large-scale systems, cloud architecture and functional programming. His published works include the highly-popular Spring Framework reference "Pro Spring".

- Personas
- **Architecture & Design**
- **Development**
- Topics
- Pivotal
- Netflix
- Spring Cloud
- Cloud
- Architecture
- Microservices

Related Editorial

Real World Microservices with Spring Cloud, Netflix OSS and Kubernetes

Moving from Monolithic Architecture to Spring Cloud and Microservices

Cloud Native Streaming and Event-driven Microservices

Architecting for Cloud Native Data: Data Microservices Done Right Using Spring Cloud

Implementing Microservices Tracing with Spring Cloud and Zipkin

Tell us what you think

Please enter a subject

Message

Allowed html: a,b,br,blockquote,i,li,pre,u,ul,p

Email me replies to any of my messages in this thread

Post Message

Community comments Watch Thread

Great Article by Nilesh Bhatt Posted Oct 18, 2016 07:50

Re: Great Article by Rob Harrop Posted Mar 04, 2017 12:11

Good article by Manu K M Posted Nov 06, 2016 11:28

Re: Good article by Rob Harrop Posted Mar 04, 2017 12:14

Nice article by Vaquar Khan Posted Mar 01, 2017 10:32

Re: Nice article by Rob Harrop Posted Mar 04, 2017 12:15

89 Shares

Article Oct 18, 2016 07:50 by "Nilesh Bhatt"

ent article!

60

you recommend proxies like NGINX along with Spring Cloud?

Reply

Back to top

article Nov 06, 2016 11:28 by "Manu K M"

ce write-up. Wondering client side load balancing can cause hitting same server instance by multiple clients since there is no centralized ement.

- Reply
- Back to top

Nice article Mar 01, 2017 10:32 by "Vaquar Khan"

Nice article , You should include API Zuul gatway in you application .

- Reply
- Back to top

Re: Great Article Mar 04, 2017 12:11 by "Rob Harrop"

Glad you liked the article. I think NGINX still has a place even with the Spring Cloud stack. It's great as an edge server for handling compression, caching, static content and some high-level routing.

- Reply
- Back to top

Re: Good article Mar 04, 2017 12:14 by "Rob Harrop"

Thanks Manu.

For sure, naive client-side balancing will exhibit that problem, but smarter algorithms can typically avoid it. A simple randomised algorithm ensures that you won't hugely overload one server, and the region/AZ-aware algorithms available with Ribbon/Eureka tend to bind client-server pairs within region/AZ. If your edge-side routing is fair, that has the effect of making your internal routing fair also,

- Reply
- Back to top

Re: Nice article Mar 04, 2017 12:15 by "Rob Harrop"

Thanks Vaquar.

I'm a huge fan of Zuul, but more recently, I've been using NGINX and the Kubernetes routing support. I think Zuul is great if you don't need to worry about integrating too closely with a container scheduler like K8S or Mesos.

- Reply
- Back to top

Close

by

on

- View
- Reply
- Back to top

Close

Quote original message

Subject Your Reply

Allowed html: a,b,br,blockquote,i,li,pre,u,ul,p

Email me replies to any of my messages in this thread

lessage Cancel

89
Shares

t Your Reply

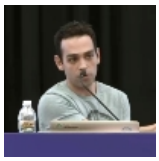
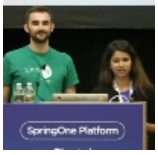
ed html: a,b,br,blockquote,i,li,pre,u,ul,p

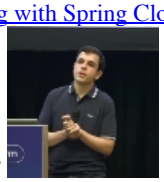
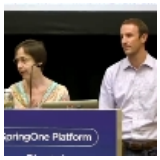
ail me replies to any of my messages in this thread

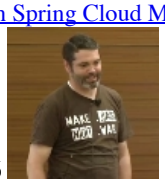
60

ncel

TED CONTENT

- Real World Microservices with Spring Cloud, Netflix OSS and Kubernetes Jan 20, 2017 
- Cloud Native Streaming and Event-driven Microservices Jan 15, 2017 
- Architecting for Cloud Native Data: Data Microservices Done Right Using Spring Cloud Jan 13, 2017 
- Implementing Microservices Tracing with Spring Cloud and Zipkin Jan 05, 2017 

- Spring Cloud on AWS Jan 01, 2017 
- Moving from Monolithic Architecture to Spring Cloud and Microservices Dec 26, 2016 

- Operating a High Velocity Large Organization with Spring Cloud Microservices Nov 30, 2016 
- Spring Cloud with Consul and Vault Dec 25, 2016 

- Leadership Election with Spring Cloud Cluster Nov 16, 2016
- Avoiding Alerts Overload from Microservices: Sarah Wells at QCon London Apr 02, 2017
- Cloud Native Computing Foundation (CNCF) Adds Linkerd, gRPC, and CoreDNS to Growing Portfolio Mar 25, 2017

- São Paulo
  Apr 24-26, 2017
- New York
  Jun 26-30, 2017
- Shanghai
  Oct 19-21, 2017
- Tokyo Fall, 2017
- San Francisco
  Nov 13-17, 2017
- London
  Mar 5-9, 2018

89 Shares

) Weekly Newsletter

ıibe to our Weekly email newsletter to follow all new content on InfoQ

60

email here Subscribe

RSS feed
For daily content and announcements
For major community updates
- For weekly community updates